Fiscal Year:	FY 2005	**Task Last Updated:**	FY 03/23/2006
PI Name:	Dietrich, Daniel L Ph.D.		
Project Title:	Development of a Portable Unit for Metabolic Analysis (PUMA)		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Other		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	44135	**Congressional District:**	10
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NOT AVAILABLE
Start Date:	11/01/2003	**End Date:**	12/31/2005
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Piltch, Nancy D. (NASA John H. Glenn Research Center) Cabrera, Marco (Case Western Reserve University) Struk, Peter M. (NASA John H. Glenn Research Center) Pettegrew, Richard D. (National Center for Microgravity Research)		
Grant/Contract No.:	IH1015113		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The objective of this project is to develop a prototype portable device that will measure human metabolic activity (Portable Unit for Metabolic Analysis, PUMA). The PUMA will provide highly time resolved measurements (relative to human respiration rate) of temperature, pressure, flow-rate, and oxygen and carbon dioxide mole fraction. The pressure, temperature and flow measurement will involve existing commercial off-the-shelf (COTS) technology repackaged into a device suitable for measurement of human metabolic activity. The oxygen partial pressure measurement will involve the modification of a COTS technology that uses the fluorescence quenching properties of oxygen. The carbon dioxide measurement will utilize the strong and relatively unique infrared absorption characteristics of the molecule to measure its concentration. The later two measurements do not necessarily involve COTS technology, but rely on well established physical principles for their measurement. All measurements will occur at the mask, and will be time resolved so that both an instantaneous and integrated signal history is possible during inhalation and exhalation. This project will exploit these advances in technology to develop a prototype portable instrument that measures important human metabolic quantities. These quantities are oxygen and carbon dioxide mole fractions, temperature, flow, and pressure during inhalation and exhalation. In contrast to existing technologies, all measurements in the proposed device will occur locally (in-situ) at the mask. The time resolution be greater than or equal to 10 Hz (possibly as high as 100 Hz). This means that it will be possible to resolve the metabolic measurements within a single breath instead of just a single integrated measure for each inhalation and exhalation.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	No progress report this reporting period.
Bibliography Type:	Description: (Last Updated: 02/27/2004)